AI & MACHINE LEARNING PRIMER: FOUNDATIONS

Chapter 1: Supervised vs Unsupervised Learning

Machine learning systems fall into three primary algorithmic paradigms:

1. Supervised Learning: Models infer a function from labeled training data containing input vectors X and target labels Y. Key applications include binary classification (e.g., medical diagnostics) and continuous regression (e.g., house price forecasting).
2. Unsupervised Learning: Discovers latent patterns, cluster boundaries, or low-dimensional representations in unlabeled data. Notable algorithms include K-Means clustering, Principal Component Analysis (PCA), and autoencoders.
3. Reinforcement Learning: Agents optimize decision policies through environmental rewards and penalties within a Markov Decision Process (MDP) setting.

ARTIFICIAL NEURAL NETWORKS & BACKPROPAGATION

Chapter 2: Deep Architectures, Activation Functions, and Optimization

Artificial neural networks consist of stacked layers of parameterized units:
- Rectified Linear Unit (ReLU): Defined as f(x) = max(0, x). Solves vanishing gradient problems across deep architectures and offers high computational efficiency.
- Sigmoid: Squashes input to the range (0, 1), standard for binary classification heads.
- Softmax: Normalizes real-valued logits into a categorical probability distribution summing to 1.0, essential for multi-class classification.

Backpropagation computes the exact analytical gradients of the loss function with respect to network weights by recursively applying the chain rule from the final layer backwards to input layers.

GENERALIZATION, OVERFITTING & REGULARIZATION

Chapter 3: Mitigating Variance and Improving Generalization

Overfitting occurs when a high-capacity model fits random noise in the training set, producing poor validation accuracy. Established regularization strategies include:

1. L1 Regularization (Lasso): Adds the absolute sum of weights to loss, enforcing parameter sparsity and serving as implicit feature selection.
2. L2 Regularization (Ridge / Weight Decay): Penalizes squared weights, ensuring smooth parameter distributions without extreme weight magnitudes.
3. Dropout: Randomly zeros out hidden unit activations during forward training passes with probability p, breaking co-adaptation between feature detectors.
4. Early Stopping: Monitors validation error and terminates training when loss consistently increases, preventing over-training.